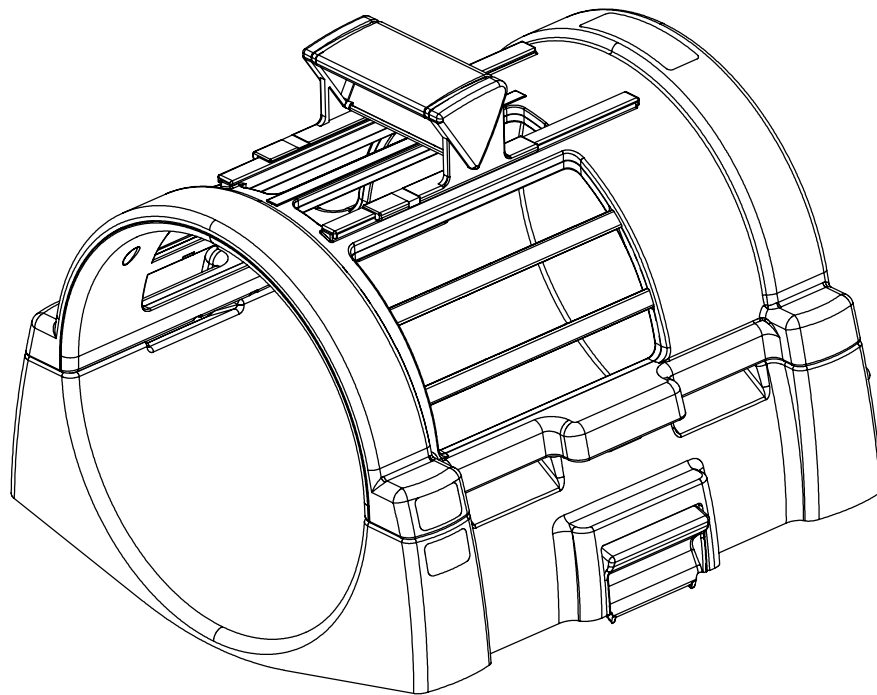


GE Medical Systems
1.0T/1.5T/3.0T Signa® Infinity/EXCITE™ Split Head Coil
Learning and Reference Guide

GE Catalog Part Number 2337543



2353925-100 Rev. 2 (01/04)



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Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:



European Representative:

GE Medical Systems Europe
Quality Assurance Manager
BP 34
F 78533 BUC CEDEX
France
Telephone: +33 1 30 70 40 40

This equipment generates, uses, and can radiate radio frequency energy. The equipment may cause radio frequency interference with other medical and non-medical devices and radio communications. To provide reasonable protection against such interference, the:

GE Signa[®] MR/i Systems

comply with emissions limits for (Group 2, Class A) Medical Devices as stated in EN 60601-1-2. However, there is no guarantee that interference will not occur in a particular installation.



If this equipment is found to cause interference (which may be determined by turning the equipment on and off), the user (or qualified service personnel) should attempt to correct the problem by one or more of the following measure(s):

- reorient or relocate the affected device(s);
- increase the separation between the equipment and the affected device;
- power the equipment from a source different from that of the affected device; and/or
- consult the point of purchase or service representative for further suggestions.

The manufacturer is not responsible for any interference caused by using interconnect cables that are not recommended or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the user's authority to operate the equipment.



Do not use devices that transmit RF Signals (**cellular phones**, transceivers, or radio controlled products) in the vicinity of this equipment as they may cause performance outside the published specifications. Keep the power to these types of devices turned off when near this equipment.

The medical staff in charge of this equipment is required to instruct technicians, patients, and other people who may be around this equipment to fully comply with the above requirement.

Immunity/Emissions Exceptions: Note the exceptions from the EMC test results. Check with the business EMC engineer for this information.

In accordance with the international safety standard IEC 601-1, this system is a Class I device, acceptable for Continuous Operation, having ordinary protection against ingress of water with type B applied parts and is not for use in the presence of flammable anesthetics.



CAUTION: User to call or contact the service representative for disposal of the MR System and its components at the end of its useful life.



CAUTION: Federal law restricts this device to sale, distribution, and use by or on the order of a physician.



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Chapter 1

About This Guide

Introduction

This chapter explains the purpose and design of this Learning and Reference Guide. It is an introduction to the guide, providing information on the purpose, prerequisite skills, guide organization, chapter format, and graphic conventions that identify the visual symbols used throughout the guide.

Safety Information

Please refer to the [Working Safely](#) chapter in this Learning and Reference Guide. The Working Safely chapter describes the safety information you and the physicians must understand thoroughly before you begin to use the system. Note that you will find additional safety information throughout your Learning and Reference Guide. If you need additional training, seek assistance from qualified GE Medical personnel.

The equipment is intended for use by qualified personnel only.

This guide should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this guide before attempting to use this product.**

Purpose of This Guide

This guide is written for health care professionals (namely, the MR technologist) to provide the necessary information relating to the proper operation of this system. The guide is intended to teach you the system components and features necessary to use it to its maximum potential. It is not intended to teach magnetic resonance imaging or to make any type of clinical diagnosis.



Prerequisite Skills

This guide is not intended to teach magnetic resonance imaging. It is necessary for you to have sufficient knowledge to competently perform the various diagnostic imaging procedures within your modality. This knowledge is gained through a variety of educational methods including clinical working experience, hospital based programs, and as part of many college and university Radiologic Technology programs.

Chapter Format

Each chapter contains a consistent format. This consistency provides uniformity for content delivery and a better learning environment for you. Listed below are the components for each chapter.

Introduction

The **Introduction** provides a short introduction to the chapter and a list of tasks to be presented.

What Do I Need to Know About...

The **What Do I Need to Know About...** section lists and explains concepts necessary to perform the tasks within the chapter.

How Do I...

The **How Do I...** section provides the detailed steps necessary to complete a given task. These detailed steps not only provide the steps to complete a task, but also provide additional information, as needed, related to a step.

Each task also includes a **Quick Steps** table. This Quick Steps table is intended to be used as a quick reference by the experienced technologist and provides only the steps necessary to complete a task. Be sure to read the detailed steps before using this table.



Mouse Controls

The mouse (Figure 1-1) is a hand-operated device that you maneuver across the surface of a pad. As you do, the on-screen cursor mimics the movement of the mouse, allowing you to move among windows and menus. For instance, moving the mouse to the right causes the on-screen cursor to move to the right. The mouse is used to make selections by clicking the left, right, and middle buttons. Table 1-1 describes the terminology used for the various mouse functions.

Figure 1-1 The Mouse

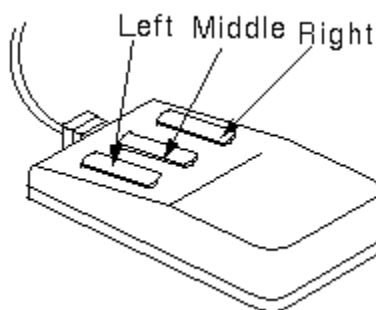


Table 1-1 Conventions for Mouse Actions

Mouse Action	Description
Click	Clicking the left mouse button to select a button or icon. The button can be pressed in, not pressed in, or popped in/out.
Right-click	Clicking the right mouse button.
Middle-click	Clicking the middle mouse button.
Click and drag	Clicking and holding the left mouse button down while dragging the cursor to the desired location.
Right-click and drag	Clicking and holding the right mouse button down while dragging the cursor to the desired location.
Middle-click and drag	Clicking and holding the middle mouse button down while dragging the cursor to the desired location.
Double-click	Clicking the left mouse button twice in rapid succession.
Triple-click	Clicking the left mouse button three times in rapid succession.



Graphic Conventions and Legends

This guide uses special conventions for graphics and legends to make it easier for you to work with the information. Table 1-2 describes the conventions used when working with menus, buttons, text boxes and keyboard keys.

Table 1-2 Conventions for Menus, Buttons, Text Boxes, and Keyboard Keys

Example	Describes
Select	Selecting an option in a check box or radial button and selecting a tab.
Press Enter	Pressing a hard key on the keyboard.
Press and hold Shift	Pressing and holding down a hard key on the keyboard.
Click [Viewer]	A button label or Interface button name.
Click  (Exam prior)	Selecting an icon-based button.
In the Matrix text box,...	The name of text box in which you can select or type text.
Type supine in the Patient Position text box (different font and bold)	Text you enter into a text box.
Select Sort > Sort by date	The pathway of selecting option(s) in a pull-down menu.



Safety Notices

The following safety notices are used to emphasize certain safety instructions. This guide uses the international symbol along with the danger, warning, or caution message. This section also describes the purpose of a Note.



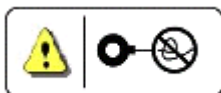
DANGER: Danger is used to identify conditions or actions for which a specific hazard is known to exist which will cause severe personal injury, death, or substantial property damage if the instructions are ignored.



WARNING: Warning is used to identify conditions or actions for which a specific hazard is known to exist which may cause severe personal injury, death, or substantial property damage if the instructions are ignored.



CAUTION: Caution is used to identify conditions or actions for which a potential hazard may exist which will or can cause minor personal injury or property damage if the instructions are ignored.



CAUTION: Coil Caution is used to identify conditions or actions for which a potential hazard of crossing or looping coil cables may exist which will or can cause minor personal injury or property damage if the instructions are ignored.

NOTE: A Note provides additional information that is helpful to you. It may emphasize certain information regarding special tools or techniques, items to check before proceeding, or factors to consider about a concept or task.



Chapter 2

Working Safely

Introduction

This guide contains detailed information on the set up, positioning, and use of your GE 1.0T, 1.5T, and 3.0T Signa® Infinity/EXCITE™ Split Head Coil. The instructions should be read carefully and thoroughly before attempting to scan patients with the coil.

This chapter is an addendum to the MR Safety Guide (Direction #2381696) or the Signa® LX™ Learning and Reference Guide Volume 1, Chapter 1, Working Safely (Direction #2274099). Those guides describe safety information you and the physician must understand thoroughly before you begin to use the system. Note that you will find additional safety information throughout your learning and reference guides. If you need additional training, seek assistance from qualified GE Medical personnel.

Make sure your Learning and Reference Guides are readily available at all times. Be sure to review the procedures and safety precautions periodically.

If you have any questions or comments on this guide, or need any assistance with the use of the product, please contact your GE Medical Systems Representative at 1-800-GE CARES.



What Do I Need to Know About...

This section presents the concepts necessary to successfully understand the safety guidelines for the GE 1.0T, 1.5T, and 3.0T Signa Infinity/EXCITE Split Head Coil. Specifically, you need to understand the following concepts:

- [Split Head Coil Safety](#)
 - [Coil Operating Safety](#)
 - [Patient Safety](#)
 - [Equipment Safety](#)
 - [Electrical and Mechanical Safety](#)
- [Emergency Procedures](#)
- [Incident Reporting](#)

Split Head Coil Safety

The Split Head Coil is intended only to be used by qualified personnel on the order of a physician. The coil is indicated for use in conjunction with a Magnetic Resonance (MR) scanner, as an accessory to produce images of the head, including but not limited to the brain, orbits, and blood vessels.

The Split Head Coil is compatible with the following Signa hardware configurations:

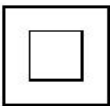
- Signa Horizon and LX Horizon 1.0T
- Signa Infinity 1.0T
- Signa Advantage 1.5T
- Signa Horizon and LX Horizon 1.5T
- Signa TwinSpeed 1.5T
- Signa Infinity 1.5T
- Signa EXCITE 1.5T
- Signa EXCITE 3.0T



Coil Classification

Product identification labels (ratings) can be used to classify the Split Head Coil and alert you to specific hazards and the level of hazard importance. Familiarize yourself with the labels in Table 2-1.

Table 2-1 Identification Labels

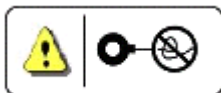
Label	Description
	Class II equivalent
	Type BF applied part

The Split Head Coil is ordinary equipment that is suitable for continuous operation.

Coil Operating Safety

Patient safety and comfort should be the primary concern during the scanning procedure. Always follow proper safety, operating, and maintenance procedures to prevent patient exposure to electrical or mechanical hazards that may cause injury.

The following general safety statements apply to scanning with an MR system. For further details, review the safety notices in your MR Safety Guide.



CAUTION: Cables should not be looped or crossed, as arcing and patient burns may result.

Route cables through the center of the magnet bore. Place cables under the cushion whenever possible. Routing near the sides of the bore increases the likelihood of cable heating from induced currents.

Keep the length of the cable in the bore to a minimum. Avoid bending the cable 180 degrees. Route the cables out of the bore in the most direct way, without looping or coiling.

Visually inspect the cable insulator jackets, strain relief and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.



CAUTION: Remove unplugged surface coils or unused accessory devices from the magnet bore; a patient burn can result.

Remove blue wire-type ECG gating cables from the magnet bore before scanning with the head coil.



CAUTION: When using any coil, make sure the patient's body does not touch the magnet bore. Use thermal resistant material or padding between the patient and the magnet to prevent burns that could be caused by patient contact with the interior of the magnet bore.

Do not allow the coil cable to touch the patient; patient burns can result. Use a thermal resistant material or pad to keep the cable from touching the patient.

Do not use coils with exposed metal conductors or damaged insulation. Skin contact with metal conductors can cause burns.

To help prevent a patient burn from closed loops formed by clasped hands, by hands touching the body, or from thighs or knees contacting over a small area, insert non-conducting pads at least 0.25 inches thick between the touching parts.



CAUTION: Use only GE or GE-authorized accessory coils and cables.

Use only accessories in good condition. If you suspect an accessory is not in good condition, discontinue its use and contact your local GE Service Engineer.

Patient Safety

The following general safety statements apply to scanning with an MR system. For further details, review the safety notices in your MR Safety Guide.



CAUTION: Monitor the patient periodically. If a patient complains of burning, tingling, stinging, or other similar burn-type sensation, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.

Ensure the patient is comfortably positioned.



CAUTION: Patients with ferromagnetic metal should not be scanned. The magnetic field may interact with implanted surgical clips or other ferromagnetic materials.

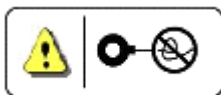
Facial makeup should be removed before scanning. It may contain metal flakes, which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.

Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing an MR examination.



CAUTION: Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.

There is a risk to scanning feverish or decompensated cardiac patients.



CAUTION: Prevent all cables from forming loops and coming into contact with the patient. Contact may result in burns to the patient.

Equipment Safety

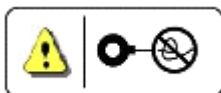
All personnel using the coil should be instructed in the proper connection and handling of the coil. Personnel should observe all safety notices that appear in this guide.



WARNING: Auxiliary equipment such as gating equipment, vital signs monitoring systems, and RF Coils that have not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of the coil.

The coil is not suitable for use in the presence of flammable anesthetics in combination with air, oxygen, or nitrous oxide.

This equipment is not protected against ingress of water.



CAUTION: Prevent cables from forming loops. Looping degrades scan performance of the coil through RF coupling.

NOTE: This equipment is rated for continuous operation.

For the disposal of this coil at the end of its useful life, please contact your GE Service Engineer.

Electrical and Mechanical Safety

The coil consists of both electrical and mechanical components. The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.



WARNING: Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

Any changes or modifications to the coil must be approved and performed by GE Medical Systems prior to installation.

Before using the coil, visually check it for any external damage. Do not use the coil if the housing or cable is broken.



CAUTION: Before installing the top half of the head coil, ensure the patient is positioned correctly to avoid potential pinch hazards to the patient. When placing the top half of the coil, you should be careful not to pinch your own fingers in between the two coil halves.

Ensure the top half of the coil is not installed backwards by locating the coil identification labels at the corners of the coil. When scanning a patient, correctly installing the top half of the head coil is paramount to receiving a good signal-to-noise ratio (SNR) from the coil.

If the head coil is not in use, it must be removed from the magnet to avoid damage.



Emergency Procedures

In the unlikely event that a coil creates smoke, sparks, or makes an unusually loud noise, or if the patient requires emergency assistance, perform these steps:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

Incident Reporting

You should contact GE Medical Systems immediately to report an incident and/or injury to a patient, operator, or maintenance employee that occurred as a result of coil operation.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

For additional information, contact:

GE Medical Systems - Americas, P.O. Box 414, Milwaukee, WI 53201

USA.....800-558-5102

Canada.....800-668-0732

GE Medical Systems - Asia

India.....91-80-845-2923

Japan.....81-120-48-2630

China.....86-21-621-92228

Australia.....61-2-9975-5501

Singapore.....65-291-8528

Taiwan.....886-2-2505-7900

GE Medical Systems - Europe

Quality Assurance Manager

BP 34

F 78533 Buc Cedex

France

Telephone: +33-1-30-70-40-40

GEMS on the World Wide Web: www.gemedicalsystems.com/rad/mri/index.html



Chapter 3

Split Head Coil

Introduction

The GE 1.0T, 1.5T, and 3.0T Signa® Infinity/EXCITE™ Split Head Coil is designed to produce images of the head, including but not limited to the brain, orbits, and blood vessels.

This chapter provides information on how to use the Split Head Coil. You should read this guide before using the coil for imaging. It also explains the step-by-step instructions to help you learn how to:

- [Perform a Quality Assurance Check](#)
- [Position the Coil](#)
- [Position the Patient](#)



What Do I Need to Know About...

This section presents the concepts necessary to successfully understand imaging with the GE 1.0T, 1.5T, and 3.0T Signa Infinity/EXCITE Split Head Coil. Specifically, you need to understand the following concepts:

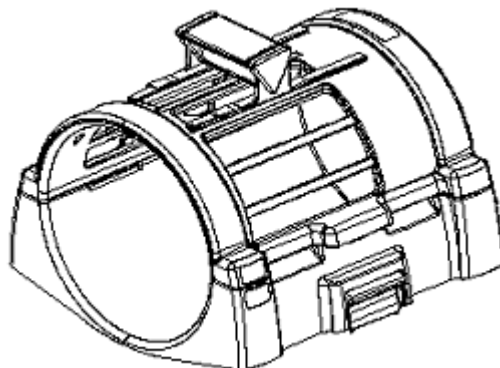
- [Split Head Coil Basics](#)
 - [Coil Components](#)
 - [Environmental Requirements](#)
 - [Coil Specifications](#)
- [Coil Setup](#)
- [Quality Assurance](#)
 - [Scanner and Coil Imaging Performance Verification](#)
 - [SNR Image Analysis](#)
 - [Cable and Hardware Checks](#)
- [Recommended Scan Protocols](#)
- [Coil Care and Maintenance](#)

Split Head Coil Basics

The Split Head Coil (Figure 3-1) is used with your Magnetic Resonance (MR) scanner, as an accessory to produce images of the head, including but not limited to the brain, orbits, and blood vessels.

The configuration and installation of your head coil requires the services of your GE Service Representative. Please contact your GE Service Representative for additional information. You may also visit GE Medical Systems on the World Wide Web at <http://www.gemedicalsystems.com/rad/mri/index.html>.

Figure 3-1 Split Head Coil





Coil Components

The coil is shipped with the head coil assembly, a disconnect box, coil adapters, cushions, and a restraint kit (Table 3-1, Table 3-2, and Table 3-3). Upon receiving your coil, check to ensure all components are included with the shipment.

Table 3-1 Shipping Components for 1.0T Systems

Description	GE Part Number	Quantity
1.0T Assembly Head Coil	2341973-2	1
1.0T Quick-Disconnect Box	2336560-2	1
Extremity Coil Adapter	2349207	1
Surface Coil Adapter	2349208	1
Split Head Coil Cushion Formed*	2330338	1
Split Head Coil Cushion Flat*	2343157	1
Head Restraint Kit	2352202	1

Table 3-2 Shipping Components for 1.5T Systems

Description	GE Part Number	Quantity
1.5T Assembly Head Coil	2341973	1
1.5T Quick-Disconnect Box	2336560	1
Extremity Coil Adapter	2349207	1
Surface Coil Adapter	2349208	1
Split Head Coil Cushion Formed*	2343157	1
Split Head Coil Cushion Flat*	2330338	1
Head Restraint Kit	2352202	1

Table 3-3 Shipping Components for 3.0T Systems

Description	GE Part Number	Quantity
3.0T Assembly Head Coil	2376114	1
3.0T Quick-Disconnect Box	2376104	1
Split Head Coil Cushion Formed*	2343157	1
Split Head Coil Cushion Flat*	2330338	1
Head Restraint Kit	2352202	1

NOTE: *Replaceable accessories



The quick-disconnect box is designed to perform the following functions:

- Isolates the radio frequency (RF) ground from the system to the head coil.
- Acts as a quadrature power splitter to drive the head coil.

The cable trap eliminates any standing waves or ground currents picked up from the environment around the head coil. An RF filter is created by resonating the ground on the cable in the quick-disconnect box to form a high impedance.

The head coil is designed to create an evenly distributed magnetic field within its volume. This field is homogeneous inside the coil and rotates in the transverse plane at the system frequency. The head coil transmits and receives a magnetic field to control the spins in the imaging volume. To do this, an oscillating magnetic field, equal to the precession frequency of hydrogen, is produced. The Larmor frequency is calculated by multiplying the gyromagnetic ratio for hydrogen (42.57 MHz/Tesla) by the static magnetic field. The field produced by the head coil must operate perpendicular (transverse) to the static magnetic field, which is applied along the z-axis.

Environmental Requirements

Store the coil in an air-conditioned scan room or equipment room. Table 3-4 describes the environmental conditions necessary for the Split Head Coil.

Table 3-4 Coil Environmental Requirements

	Transport and Storage	Operating Conditions
Barometric Pressure	7.25 to 15.374 PSI (500 hPa to 1060 hPa)	10.15 to 15.374 PSI (700 hPa to 1060 hPa)
Humidity	10% to 100% Non-condensing	30% to 75% Non-condensing
Temperature	-40°F to 158°F (-40°C to 70°C)	41°F to 104°F (5°C to 40°C)

NOTE: PSI = Pounds per square inch

hPa = hecta Pascals

Coil Specifications

The design specifications for the Split Head Coil are defined in Table 3-5.

Table 3-5 Coil Specifications

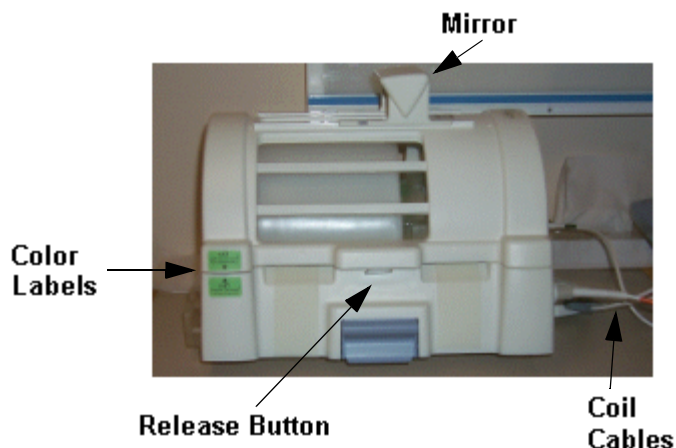
Field Strength	Coil Model	GE Part Number	Frequency
1.0T	Split Head Coil	2341973-2	42.68 MHz
1.5T	Split Head Coil	2341973	63.86 MHz
3.0T	Split Head Coil	2376114	127.72 MHz



Coil Setup

The Split Head Coil (Figure 3-2) sits on the imaging table so the cable is pointing into the magnet bore. The top of the coil should be connected so that the color labels on the top and base of the coil match. Image quality is compromised if the labels are on opposite sides of the coil. In addition, the mirror will be on backwards if the coil halves are not properly aligned. The mirror on the top of the coil slides, but does not lock. The top half of the coil can be removed by pressing up on the release buttons on the sides of the coil.

Figure 3-2 Split Head Coil



There are connectors on the base of the coil that snap into position with the top half of the coil. The red male parts receive the blue female parts (Figure 3-3).

Figure 3-3 Coil Connectors



There are Velcro® straps (Figure 3-4) that attach directly onto the coil, with two Velcro receivers on either side of the blue table connector. There are two chin straps and two wide, forehead straps.

Figure 3-4 Velcro Straps





Quality Assurance

A Quality Assurance (QA) check is used to check the coil for image integrity. The [Perform a Quality Assurance Check](#) procedure, described in the **How Do I...** section of this guide, should be completed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

Scanner and Coil Imaging Performance Verification

Your service engineer should perform system level Signal-to-Noise Check prior to your use of the coil. They should refer to the Service Methods CD for additional information.

The Split Head Coil can only be used in one mode of operation. It can be selected in the Patient Position area by clicking the button to the right of the **Coil** text box and selecting **Head** on the Coil Names window.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is a Fast Spin Echo-Excel (FSE-XL) sequence used to minimize susceptibility and inhomogeneity effects. The noise scan is a Gradient Echo (GRE) sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan must be run prior to the noise scan; the Scan TR (R1 and R2) and Transmit Gain (TG) values from the signal scan are used when acquiring the noise scan. An Auto Prescan should not be run prior to the noise scan, as the values change.

NOTE: Refer to [QA References](#) in the appendix for the signal and noise protocols.

Figure 3-5 displays the Head SNR Phantom (GE part number for 1.0T/1.5T - 2321556, GE part number for 3.0T - 2371512) required to complete the verification of the coil imaging performance. This special phantom cannot be used on other coils, nor should other coil phantoms be used with this coil. The phantom should be placed in the coil so that it fits snugly up against the coil front.

Figure 3-5 Head SNR Phantom

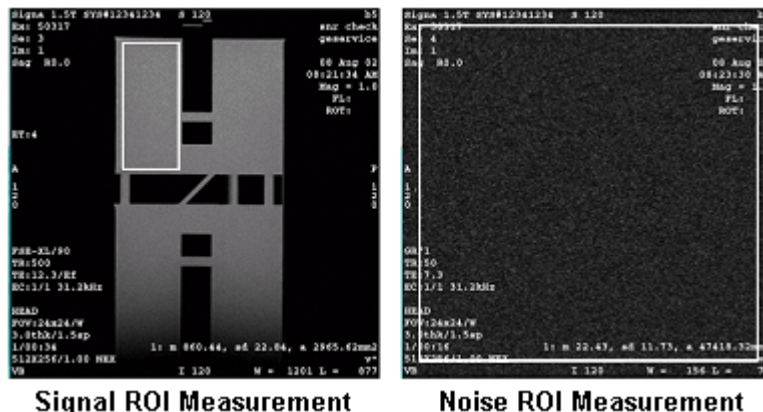




SNR Image Analysis

Since a significant decrease in the SNR may not be visually obvious on a phantom image, perform a region of interest (ROI) check. A rectangular ROI should be used for the signal and noise measurements. All ROI measurements are made on the individual element images, not on the composite image. The actual ROIs should be similar in shape and size to those in Figure 3-6.

Figure 3-6 Rectangular ROIs



ROIs in both signal and noise images can be measured directly in the Viewer. Use the measure feature to select a rectangular shape ROI, and adjust its size and orientation so the ROI is similar to those shown in Figure 3-6. Mean, standard deviation, and area of the ROI appear in the lower right corner of the image.

Record these values on the [SNR Data Sheet](#) (Table in Appendix 1). Compare this SNR data with the last acquired data.

Individual receiver SNR is defined as the mean of data within the signal ROI divided by the standard deviation of data within the noise ROI. The SNR calculation (Equation 3-1) uses the mean of the signal image and the mean of the noise image.

Equation 3-1 SNR Equation

$$\text{SNR}_{\text{Element}} = \frac{\text{Mean of Signal within ROI}}{(1.253) \times (\text{Mean of Noise within ROI})}$$

The SNR measurements must be greater than or equal to the specifications in Table 3-6.

Table 3-6 SNR Specifications

Coil	SNR
1.0T Split Head Coil	18.0
1.5T Split Head Coil	29.1
3.0T Split Head Coil	21.8



Cable and Hardware Checks

Due to pin damage concerns, no cable continuity check is recommended for the connector. However, check the cables for mechanical damage and shorted center pins. Also inspect the main pins for corrosion and proper mechanical operation.

You should routinely visually inspect the connector for damage from improper insertion. Banana plugs damage pins if you attempt to insert the connector upside down. At this time, also inspect the connector for foreign material and remove as appropriate.

While completing these checks, make sure the cradle latches and top latches are operating and have adequate range of motion.

Recommended Scan Protocols

Use the system recommended protocols for scanning with the Split Head Coil for optimal results and best performance. The specific choice of protocols is entirely the responsibility of you or your radiologist.

Coil Care and Maintenance

The Split Head Coil should be stored in an air-conditioned scan or equipment room. To store the coil, a storage space of 420 mm x 430 mm x 368 mm is required.



WARNING: Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.



CAUTION: The coil and patient comfort pads must be cleaned and stored using the following procedure:

1. Wipe the coil and pads clean with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
2. Do not spray or pour any cleaning solution directly on the coil!
3. Let the coil housing and pads dry before use.
4. Store the coil in an air-conditioned scan room or equipment room.
5. Clean the coil and pads once a day, or as required.

Under no circumstances should the coil be placed into any type of sterilizer!



How Do I...

This section provides the step-by-step instructions for using the Split Head Coil. Specifically, it describes how to perform the following procedures:

- Perform a Quality Assurance Check
 - [Position the Coil and Phantom](#)
 - [Acquire a Signal Scan](#)
 - [Acquire a Noise Scan](#)
 - [Measure the SNR](#)
- [Position the Coil](#)
- [Position the Patient](#)



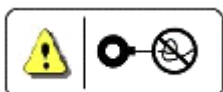
Perform a Quality Assurance Check

Position the Coil and Phantom

The Quality Assurance (QA) check should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

Use this procedure to position the Split Head Coil for a QA check.

1. Remove any other coil or unused accessory device from the magnet.



CAUTION: Patient burns may result from coil to coil coupling.

2. Position the Split Head Coil on the patient table.
 - a) Place the posterior half of the coil at the top of the patient cradle.
 - b) Make sure the cables extend into the magnet.
 - The cables from the quick-disconnect box to the head coil should always be straight, with no crossovers or loops evident from the outside.
 - c) Lock the baseplate into the table slots.
3. Unlock and remove the top half of the coil by depressing the latches on either side of the coil.
4. Remove any pads from the coil.
5. Mount the phantom onto the phantom positioner and place on the phantom pad.
6. Slide the pad, positioner, and phantom onto the coil base.
7. Reattach the top half of the coil.
8. Slide the phantom positioner so the landmarks on the phantom positioner align with those on the coil.
9. At the magnet, press **Alignment Light** to turn on the light.
10. Move the cradle to align the coil with the alignment lights.
11. Press **Landmark** to communicate the region of interest center to the system.
12. Press **Move to Scan** to move the coil to scan position.
 - ♦ Ensure the coil cable does not get snagged.





Quick Steps: Perform a Quality Assurance Check – Position the Coil and Phantom

1. Remove any other coil or unused accessory device from the magnet.
2. Position the Split Head Coil on the patient table.
3. Unlock and remove the top half of the coil by depressing the latches on either side of the coil.
4. Remove any pads from the coil.
5. Mount the phantom onto the phantom positioner and place on the phantom pad.
6. Slide the pad, positioner, and phantom onto the coil base.
7. Reattach the top half of the coil.
8. Slide the phantom positioner so the landmarks on the phantom positioner align with those on the coil.
9. At the magnet, press Alignment Light to turn on the light.
10. Move the cradle to align the coil with the alignment lights.
11. Press Landmark to communicate the region of interest center to the system.
12. Press Move to Scan to move the coil to scan position.



Perform a Quality Assurance Check

Acquire a Signal Scan

The signal scan is an FSE-XL sequence used to minimize susceptibility and inhomogeneity effects. The signal scan must be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

Use this procedure to acquire a signal scan for a QA check.

1. Click the **Scan Rx Desktop** icon on the control panel.
2. Click **[New Pt]** in the Patient Register area.
 - ♦ The Patient Information area appears.
3. Complete the **Patient ID**, **Patient Name**, and **Weight** text boxes.
 - a) Enter **geservice** in the **Patient ID** text box.
 - b) Enter **snr check** in the **Patient Name** text box.
 - c) Enter **111** in the **Weight** text box.
4. Click **[Patient Position]**.
5. Enter the parameters from the Signal protocol.
 - ♦ Refer to [QA References](#) in the appendix for the [Signal and Noise Protocols](#).
6. Click **[Save Series]** to close the current prescription window.
7. Click **[Prepare to Scan]** to download the acquisition.
8. Click **[Auto Prescan]**.
9. Take note of the R1, R2, and TG values on the SNR data sheet.
 - ♦ Refer to [QA References](#) in the appendix for the [SNR Data Sheet](#).
 - ♦ R1 should equal 11, and R2 should equal 14. If not, run Manual Prescan and adjust the R1 and R2 values.
10. Click **[Scan]**.



Quick Steps: Perform a Quality Assurance Check – Acquire a Signal Scan

1. Click the Scan Rx Desktop icon on the control panel.
2. Click [New Pt] in the Patient Register area.
3. Complete the Patient ID, Patient Name, and Weight text boxes.
4. Click [Patient Position].
5. Enter the parameters from the Signal protocol.
6. Click [Save Series] to close the current prescription window.
7. Click [Prepare to Scan] to download the acquisition.
8. Click [Auto Prescan].
9. Take note of the R1, R2, and TG values on the SNR data sheet.
10. Click [Scan].



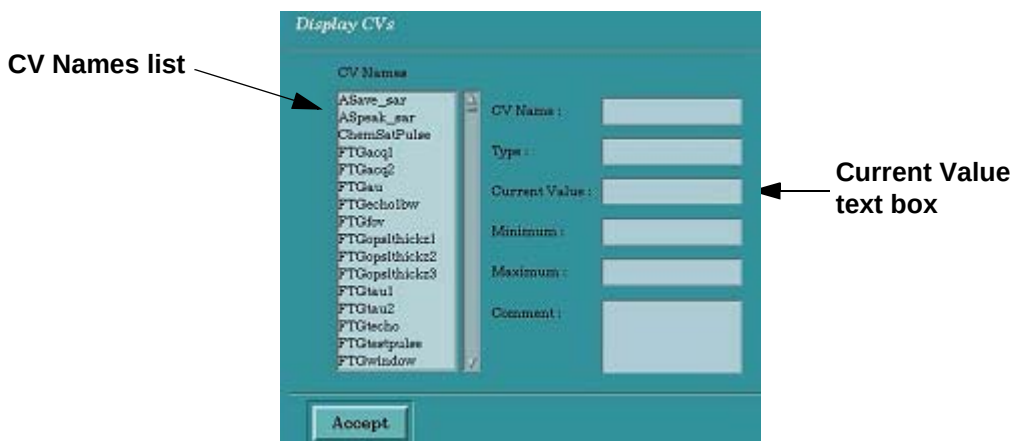
Perform a Quality Assurance Check

Acquire a Noise Scan

The noise scan is a GRE sequence that uses a Control Variable to eliminate the transmit RF completely during the scan. A signal scan must be run prior to the noise scan as the same R1, R2, and TG values must be used for both the signal and noise scans. Do not run an Auto Prescan prior to the noise scan as the values will be changed.

Use this procedure to acquire a noise scan for a QA check.

1. Copy the signal scan series.
 - a) Select the signal series in the Rx Manager.
 - b) Right-click and drag to select **Paste**.
2. Click **[View Edit]**.
 - ♦ This allows you to edit the parameters of the copied series.
3. Enter the parameters from the Noise protocol.
 - ♦ Refer to [QA References](#) in the appendix for the [Signal and Noise Protocols](#).
4. Click **[Save Series]** to close the current prescription window.
5. Click **[Prepare to Scan]** to download the acquisition.
6. Right-click **[Research Operations...]** and select **User CVs**.
 - ♦ The Display CVs window opens.



7. Double-click **rhformat** in the CV Names list.
 - ♦ Rhformat is entered in the **CV Name** text box.
 - ♦ Alternatively, you may type the name in the text box.
8. Enter 1 in the **Current Value** text box.



9. Double-click **do_noise** in the CV Names list.
 - ♦ do_noise is entered in the **CV Name** text box.
 - ♦ Alternatively, you may type the name in the text box.
10. Enter 1 in the **Current Value** text box.
 - ♦ Turning this value on eliminates the transmit RF completely during the scan.
11. Click **[Accept]**.
 - ♦ The Display CVs window closes.
12. Click **[Manual Prescan]**.
 - ♦ Do not make any changes.
13. Click **[Done]**.
 - ♦ The Manual Prescan window closes.
14. Click **[Scan]**.

Quick Steps: Perform a Quality Assurance Check – Acquire a Noise Scan

1. Copy the signal scan series.
2. Click [View Edit].
3. Enter the parameters from the Noise protocol.
4. Click [Save Series] to close the current prescription window.
5. Click [Prepare to Scan] to download the acquisition.
6. Right-click [Research Operations...] and select User CVs.
7. Double-click rhformat in the CV Names list.
8. Enter 1 in the Current Value text box.
9. Double-click do_noise in the CV Names list.
10. Enter 1 in the Current Value text box.
11. Click [Accept].
12. Click [Manual Prescan].
13. Click [Done].
14. Click [Scan].



Perform a Quality Assurance Check

Measure the SNR

A rectangular ROI should be used to measure the signal and noise on your acquired images. Both the signal and noise images can be measured in the Viewer.

Use this procedure to measure the SNR on your signal and noise images.

1. Select the signal image in the Browser and click **[Viewer]**.
 - ♦ The Viewer displays the signal image.
2. Click **[Measure]**.
 - ♦ A menu appears, displaying available measurement methods.
3. Click the **Rectangular ROI** icon.
 - ♦ A rectangular ROI is deposited on the center of the image.
4. Click and drag the small box to resize the ROI.
 - ♦ The orientation of the ROI should be similar to the following figure.
 - ♦ The mean, standard deviation, and area of the ROI appear in the lower right corner of the image.
5. Record the values for the mean signal on the [SNR Data Sheet](#).
 - ♦ Refer to [QA References](#) in the appendix.
6. Repeat steps 1 through 5 with the noise image, recording the mean value for the noise signal.
 - ♦ The orientation of the noise ROI should be similar to the following figure.





Quick Steps: Perform a Quality Assurance Check - Measure the SNR

1. Select the signal image in the Browser and click [Viewer].
2. Click [Measure].
3. Click the Rectangular ROI icon.
4. Click and drag the small box to resize the ROI.
5. Record the values for the mean signal on the SNR Data Sheet.
6. Repeat steps 1 through 5 with the noise image, recording the mean value for the noise signal.

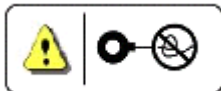


Position the Coil

The Split Head Coil is a transmit and receive coil designed to create an evenly distributed magnetic field within its volume. You should complete the quality assurance procedure before using the coil for patient imaging.

Use this procedure to position the Split Head Coil for patient scanning.

1. Remove any other coil or unused accessory device from the magnet.



CAUTION: Patient burns may result from coil to coil coupling.

2. Position the Split Head Coil on the patient table.
 - a) Place the posterior half of the coil at the top of the patient cradle.
 - b) Make sure the cables extend into the magnet.
 - The cables from the quick-disconnect box to the head coil should always be straight, with no crossovers or loops evident from the outside.
 - c) Lock the baseplate into the table slots.
3. Place the comfort pads into the coil.
 - ♦ The large pads should be placed on the cradle so the patient is positioned comfortably on the table with his or her head centered in the head coil.
4. Place a clean cotton sheet over the coil and comfort pad.
 - ♦ Cover these items so the patient's skin will not come in contact with the coil or the comfort pad.

Quick Steps: Position the Coil

1. Remove any other coil or unused accessory device from the magnet.
2. Position the Split Head Coil on the patient table.
3. Place the comfort pads into the coil.
4. Place a clean cotton sheet over the coil and comfort pad.



Position the Patient

The Split Head Coil is a transmit and receive coil used to produce images of the brain, orbits, and blood vessels. It may also be used for any body part that fits into the coil, but remember to place sponges so the patient's skin does not touch the coil.

Use this procedure to position your patient for a scan using the Split Head Coil.

NOTE: You must complete the Position the Coil procedure before starting this procedure.

1. Position the patient head-first and supine, and place the head as far into the coil form as possible.
 - ♦ The patient's head should be in the posterior half of the coil.
 - ♦ The patient's shoulders should be against the inferior edge of the coil.
2. Instruct the patient to place his or her arms at his or her sides during the examination.
 - ♦ Do not allow the patient's arms to touch the bore. Place sponges between the patient's arms and the bore if needed.
 - ♦ Do not allow the patient to cross his or her arms.
3. Use additional pads to immobilize and make the patient comfortable.
4. Provide the patient with the Patient Alert Bulb and give instructions on its use.
5. Provide ear plugs for the patient after all instructions have been given.



WARNING: Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

6. Place the top half of the coil over the patient's head and close the latches.
7. Plug the coil connector into the head port of the carriage.
 - ♦ Ensure the coil cable is not looping or crossed at any point within the magnet bore.
8. Instruct the patient to close his or her eyes.



CAUTION: Instruct the patient to close his or her eyes before turning on the alignment light and landmarking. The eyes must be protected from laser radiation. Exposing eyes to the laser alignment lights may result in eye injury.



9. Press **Alignment Light** to turn on the light.
10. Move the table to align the light at the center of the coil, approximately at the patient's nasion.
 - ♦ The localizing light should be aligned with the cross hairs on the coil for positioning accuracy within the bore.
11. Press **Landmark** to communicate the region of interest center to the system.
12. Press **Move to Scan** to move the patient to the isocenter.
13. Proceed with the scan prescription, selecting **Head** for your coil selection.

Quick Steps: Position the Patient

1. Position the patient head-first and supine, and place the head as far into the coil form as possible.
2. Instruct the patient to place his or her arms at his or her sides during the examination.
3. Use additional pads to immobilize and make the patient comfortable.
4. Provide the patient with the Patient Alert Bulb and give instructions on its use.
5. Provide ear plugs for the patient after all instructions have been given.
6. Place the top half of the coil over the patient's head and close the latches.
7. Plug the coil connector into the head port of the carriage.
8. Instruct the patient to close his or her eyes.
9. Press Alignment Light to turn on the light.
10. Move the table to align the light at the center of the coil, approximately at the patient's nasion.
11. Press Landmark to communicate the region of interest center to the system.
12. Press Move to Scan to move the patient to the isocenter.
13. Proceed with the scan prescription, selecting Head for your coil selection.



Appendix A

QA References

Introduction

This appendix includes the quality assurance (QA) references necessary to perform the signal-to-noise-ratio (SNR) checks on your GE 1.0T, 1.5T, or 3.0T Signa® Infinity/EXCITE™ Split Head Coil. Specifically, it includes the protocols and a sample recording data sheet. Keep the completed data sheets with your system data sheets for future reference.



Signal and Noise Protocols

Protocol	Signal	Noise
Patient Exam Information		
Patient ID	geservice	geservice
Patient Name	Head	Head
Patient Weight	111 lbs (50 kg)	111 lbs (50 kg)
Patient Position		
Patient Position	Supine	Supine
Patient Entry	Head First	Head First
Coil	Head	Head
Series Description	Signal	Noise
Imaging Parameters		
Plane	Axial	Axial
Mode	2D	2D
Pulse Sequence	FSE-XL	Gradient Echo
Imaging Options	None	None
Scan Timing		
# of Echoes	1	1
TE	17	Min Full
TR	500	50
Flip Angle	N/A	1
Echo Train Length	4	N/A
Bandwidth	31.25	31.25
Acquisition Timing		
Frequency	512	512
Phase	256	256
NEX	1	1
Phase FOV	1	1
Frequency Direction	A/P	A/P
Auto Center Frequency	Peak	Peak
Autoshim	On	On
Phase Correct	On	N/A
Scanning Range		
FOV	24	24
Slice Thickness	3	3
Spacing	1.5	1.5
Start R/L, P/A Center, I/S Center, End R/L	0.0	0.0
Slices	1	1
Table Delta	0.0	0.0

[illegible]



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